FRUIT QUALITY PREDICTION

Base Paper vs Proposed Project

Complete step-by-step workflows using one realistic mango-orchard example

Purpose: Explain what enters each system, what happens at every stage, which technology performs the stage, and what the final output means. The numbers used in the walkthrough are illustrative examples—not results reported by the paper.

Base paper: Kim & Sun, “Balanced Multi-Source Transfer Learning for Fruit Quality Prediction With Limited Data,” IEEE Access (2026), DOI: 10.1109/ACCESS.2026.3697751.

Proposed project: “A Digital Twin-Driven Multimodal Framework for Early Fruit Quality Prediction Using Vision-Language Models.”

Quick idea before the details

Question	Base paper	Proposed project
When does it predict?	Uses labelled crop-quality records and associated growing/weather features; it is not an early-image pipeline.	Aims to predict before harvest from early flower/leaf images plus plant
Main problem solved	How to use other crop datasets without letting large source datasets overpower a small target-crop dataset.	How to combine different kinds of early evidence for plant-level quality
Main technologies	BMTL/LOO-Select + tabular regressors such as Ridge; preprocessing and validation.	ViT/CNN image encoder + tabular model + fusion; plant-level digital-tw
Typical output	Numeric regression value, such as predicted Brix, evaluated with R²/MAE.	Several plant-level estimates, uncertainty, contributing evidence, harv

PART 1 — Base paper workflow

The base paper is primarily a data-balancing and transfer-learning method. It does not take a photograph of a flower and send it through a Vision Transformer. Its key question is: “If I have only a small amount of labelled data for my target crop, can I borrow useful labelled data from other crops without letting those larger datasets dominate?”

Example scenario (illustrative)

Imagine a research team wants to predict the sweetness of apple samples, measured in degrees Brix (°Brix). They have relatively few labelled apple records, while other fruit datasets—such as citrus, grape and pear—have different sample counts. Each historical record contains structured values such as fruit measurements, crop/variety information and weather summaries.

Important: the example below explains the algorithm. The exact record values and model predictions are made-up for teaching; they are not the paper’s raw rows or reported predictions.

STEP 1 — Collect labelled historical records

Input: rows of structured data. Example row: crop=apple; variety=A; region=orchard zone 1; growing-degree-days=1,250; average temperature=24°C; rainfall=42 mm; fruit weight=160 g; measured Brix=12.4. Other rows come from citrus, grape and pear. Technology: CSV/data tables, Python/pandas or equivalent preprocessing. The measured Brix is the label the model learns to predict.

STEP 2 — Clean and standardize the columns

The team checks missing values, consistent units and categories. Numeric inputs are standardized using z-scores; the paper also reports PCA retaining 95% of variance and uses domain-distance analysis such as MMD. Why: a feature measured in hundreds should not dominate a feature measured in decimals simply because its numeric scale is larger.

STEP 3 — Separate a small target-crop set

Apple is the target. The apple records are split into target-training and target-validation portions (the paper describes an 80/20 split). Validation records are held back for estimating how well each source crop transfers to apples. Technology: controlled train/validation split. The held-out labels must not be used to fit the candidate source model.

STEP 4 — Train one candidate model per source crop

Train a regressor on citrus records, another on grape records, and another on pear records. The paper's primary learner is Ridge regression (alpha=1.0); Gradient Boosting, Random Forest and RBF-SVR are also evaluated. Each source model is then tested on the same held-out apple validation records.

STEP 5 — Measure how useful each source is for apples

Suppose, purely as an example, the validation MAEs are: citrus model=1.0 °Brix; grape model=0.7 °Brix; pear model=1.4 °Brix. Lower MAE means smaller average prediction error on the apple validation set. Therefore grape looks most transferable in this invented example, citrus next, and pear least useful. Technology: MAE evaluation.

STEP 6 — Convert errors into source weights

BMTL uses inverse-error-style transferability scores and normalizes them into weights. In plain language: sources that predict the target validation samples more accurately receive more influence; sources that do poorly receive less. The numbers above are only illustrative, and the actual normalized weights depend on the implementation.

STEP 7 — Choose a source/target balance

The method selects a controlled balance ratio, with candidate values such as 0.3, 0.5 and 0.8 tested in the paper. It then samples a controlled number of source rows according to their weights, instead of blindly combining every available row. Purpose: stop a large source dataset from overwhelming the small apple dataset.

STEP 8 — Train the final balanced regressor

Combine the apple training rows with the selected and weighted/sampled source rows. Fit the final regression model on this balanced training set. If the target set is extremely small (the paper discusses nT<50), its LOO-Select strategy may choose one best source rather than mix all sources.

STEP 9 — Predict on unseen apple records

Input: a new apple record with the permitted structured features, such as weather/growing-degree-days and other fields used by the trained model. Output: a numeric estimate, for example 13.1 °Brix (illustrative only). Compare it with the measured Brix later. The system does not automatically create a farmer-facing explanation or a plant photo analysis.

STEP 10 — Evaluate and decide whether transfer helped

Compare BMTL against target-only, naive pooling and other baselines using held-out data and metrics such as R^2 and MAE. If target-only performs better, do not assume transfer is beneficial. The paper reports an apple case improvement from R^2=0.393 to R^2=0.536 in one comparison, but also reports cases where BMTL underperforms target-only.

Base paper flow in one line

Historical labelled crop tables → clean/standardize → split target records → train source-specific regressors → score sources on target validation → weight/sample source records → train balanced final regressor → predict numeric quality → evaluate against held-out labels.

What is special about it? BMTL changes which training rows and how many source rows are used. It is not itself an image model or a digital twin.

PART 2 — Proposed project workflow

Your proposed project changes the timing and the information sources. The goal is to estimate fruit quality while the fruit is still developing, using early-stage images and the plant's environment/management history, then maintain a plant-level record and explain the prediction.

Example scenario (illustrative)

Imagine a mango orchard with plant ID MG-042. The mangoes are not ready for harvest. A grower captures flower and leaf photos, and the app has the plant's variety, age, soil test, local weather history and irrigation log. Later, at harvest, the team measures Brix, weight and firmness to create training labels. The example output below is hypothetical and must not be presented as validated model performance.

STEP 1 — Register the individual plant

Input: plant ID MG-042; crop=mango; variety=Alphonso (example); planting date; orchard/plot location; soil-test ID. Technology: React form + Spring Boot REST API (if used) + MySQL plant registry. The system creates a stable plant identity so photos, weather, management and later harvest measurements all connect to the same plant.

STEP 2 — Capture early flower and leaf images

Input: dated photographs of flowers and leaves, captured before the fruit is mature. Store plant_id, date, growth stage and image type with every file. Technology: mobile/web camera upload and image storage. These photos are not treated as ground-truth quality labels; they are visual evidence the model may learn to associate with later outcomes.

STEP 3 — Extract visual features

The image branch resizes/normalizes the photos and feeds them to a pretrained Vision Transformer (ViT) or a CNN baseline using PyTorch. It produces a numerical image embedding—features learned from the visual patterns. ViT is a proposed architecture from your abstract; a CNN baseline is recommended for fair comparison.

STEP 4 — Collect soil, weather and management inputs

For MG-042, inputs might include a measured soil pH/nutrient panel, daily temperature/rainfall/humidity, irrigation dates and amounts, fertilizer records, variety and plant age. Technology: local soil test, NASA POWER weather data where appropriate, farm logbook/database, pandas for joining tables. Values must be real records, not invented defaults.

STEP 5 — Align every input to the prediction date

Suppose the grower asks for a prediction on 15 March. The system only uses flower/leaf photos and weather/management data available on or before 15 March. It must not use a later fruit photo or the eventual harvest Brix as an input. Technology: timestamp joins and data-quality checks. Split training/testing by plant, plot or season to reduce leakage.

STEP 6 — Build tabular features

Convert weather history into documented summaries, such as recent average/min/max temperature, rainfall and humidity; add soil, variety, plant age and irrigation summaries. Technology: pandas + scikit-learn preprocessing; a CatBoost, Random Forest or Gradient Boosting model can form the tabular-only baseline.

STEP 7 — Fuse image and tabular evidence

The ViT image embedding is combined with the normalized tabular features. A first implementation can concatenate both feature vectors and train a prediction head; later experiments can compare late fusion or gated/attention fusion. Technology: PyTorch multimodal model. Compare image-only, tabular-only and fused versions to prove whether combining modalities helps.

STEP 8 — Use a graph only if relationships are real

Your abstract names a GNN. A meaningful graph might link plant nodes to plots, nearby weather stations and management events. Technology: PyTorch Geometric. Do not add a GNN just for appearance: if the dataset has no defensible nodes/edges or enough examples, begin with the fusion model and treat the GNN as an extension.

STEP 9 — Predict later fruit outcomes from early evidence

The trained model outputs estimated Brix, fruit weight, firmness and expected harvest date, but only for targets with enough reliable measured training labels. Example hypothetical output for MG-042: Brix=12.8 °Brix; expected harvest window=20–27 June; predicted fruit weight=310 g; firmness=moderate/high on a defined scale. These values are illustrative—not actual predictions or promises.

STEP 10 — Estimate uncertainty and contributing evidence

The app should show a prediction range or uncertainty estimate if it has been calibrated and validated. SHAP can summarize tabular feature contributions; image saliency can indicate regions influential to the model, but saliency is not causal proof. If uncertainty is high or inputs are missing, show that limitation rather than presenting a precise-looking answer.

STEP 11 — Generate a grounded farmer explanation

Pass the structured outputs, uncertainty and verified attributions to a VLM/language layer. Example: “For plant MG-042, the model estimates 12.8 °Brix, with a predicted harvest window of 20–27 June. The estimate is influenced by the recorded recent weather and irrigation features. Confidence is limited because this farm/variety has few labelled examples.” The text must only describe evidence supplied to it; it must not invent a disease, nutrient deficiency or causal claim.

STEP 12 — Update the digital twin and dashboard

Save the input snapshot, image references, prediction, model version, uncertainty, explanation and date in the plant’s history. The React dashboard displays the plant timeline and later compares predictions with actual harvest measurements. Technology: React + REST API (Spring Boot or FastAPI) + MySQL; Python FastAPI can serve model inference, with Spring Boot calling it if that is the chosen backend.

STEP 13 — Measure the real result and retrain carefully

At harvest, collect the actual Brix with a refractometer, fruit weight, firmness and harvest date using a consistent protocol. Link them to MG-042. Evaluate MAE/RMSE/R^2 and harvest-date error on plants/seasons never used in training. Add new records to future training only through a documented versioned pipeline.

Proposed project flow in one line

Register plant → capture dated early flower/leaf photos → join weather/soil/irrigation/variety/age → extract image features with ViT + tabular features → fuse features → predict quality and harvest timing → estimate uncertainty/attributions → generate evidence-grounded VLM explanation → store plant history in digital twin → compare with measured harvest outcomes.

PART 3 — Direct comparison of outputs

Aspect	Base paper	Proposed project
Input example	Structured labelled crop records plus weather/growing features; other crop records are sources.	MG-042 flower/leaf photos + variety/age + soil + weather + irrigation/man
Core processing	BMTL estimates which source crops transfer well and balances samples against target rows; a tabular regressor	ViT/CNN extracts image embeddings; tabular model encodes environment/
Example output	One numeric target estimate, e.g. 13.1 °Brix in the teaching example.	Plant-level multi-output estimates, e.g. 12.8 °Brix, 310 g, harvest window 2
What the user sees	Research evaluation of predicted values versus known labels, primarily metric tables.	A dashboard for each plant: estimates, evidence, uncertainty, explanation,
Main contribution	Control source/target imbalance to reduce negative transfer when labelled target data is limited.	Early-stage multimodal prediction and plant-level tracking/explanation—no
Data challenge	Paper’s Korean multi-crop dataset is described as available upon reasonable request.	Requires linked early images, time-aligned environment/management, and

What is different, in the simplest words?

- Base paper: “I already have structured historical measurements from several crop types. Which other-crop records should I use, and in what balance, to improve a prediction for my target crop?”

- Proposed project: “Before this plant’s fruit is ready, I have photos and growing-condition records. What might its fruit quality and harvest timing be, how uncertain is that estimate, and how can I explain it to the grower?”

- BMTL and multimodal fusion solve different problems. They can potentially be combined later: a multi-crop training strategy could help when you have several crop/variety datasets, while the image-and-tabular fusion model handles the mixed input types.

- A VLM does not make the numerical prediction reliable by itself. The prediction model must be trained and validated on measured outcomes; the VLM should explain those outputs without inventing evidence.
- Shelf life should remain optional until you can collect repeated observations under a defined storage protocol. Start with one crop, Brix, weight and harvest date if resources are limited.

Implementation order recommended

Phase	Build this first	Why
1	One crop + plant IDs + early photos + harvest labels	Without paired labels, supervised early-quality prediction cannot be validat
2	Tabular-only, image-only and fused baselines	Shows whether each modality and fusion actually improve prediction.
3	Uncertainty + SHAP/image evidence + digital-twin history	Makes the result inspectable and tracks what was known at prediction time
4	Grounded VLM explanation and dashboard polish	Adds useful natural-language communication after numerical outputs are d
5	GNN and BMTL transfer experiments	Add only if the graph structure and multi-crop labelled data justify the comp

Source boundary and cautions

The base-paper description and reported numerical results in this guide follow the uploaded paper and the previous project analysis. The workflow architecture for the proposed project is a recommendation based on your abstract. The mango example, its plant ID, input values and predicted outputs are invented for teaching only; they are not observations, measured results or performance claims.

Base paper reference: Kim, Y. & Sun, E. J. (2026), “Balanced Multi-Source Transfer Learning for Fruit Quality Prediction With Limited Data,” IEEE Access. DOI: https://doi.org/10.1109/ACCESS.2026.3697751.

Potential data resources: PlantVillage (leaf-health images, not paired fruit-quality labels): https://data.mendeley.com/datasets/tywbtsjrjv/1 ; Fruits-360 (mostly harvested fruit images): https://github.com/fruits-360 ; NASA POWER daily weather: https://power.larc.nasa.gov/docs/services/api/temporal/daily/ ; FAO Soils Portal: https://www.fao.org/soils-portal/ . Verify access, suitability and licensing before use.